THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET D

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An event occurred 6 times in 40 trials. Its experimental probability (relative frequency) is: 1

(a) 0.15 **(b)** 6.67 **(c)** 0.25 **(d)** 0.075

2. The number of outcomes in the sample space when a coin is tossed and a die is rolled is: 1

(a) 6 **(b)** 24 **(c)** 14 **(d)** 12

3. A sample shows a probability of $\frac{3}{20}$ that a student likes cricket. In a school of 800 students the expected number who like cricket is: 1

(a) 120 **(b)** 60 **(c)** 220 **(d)** 200

4. If $P(E) = 0.35$, then the probability that E does not happen is: 1

(a) 1.35 **(b)** 0.35 **(c)** 0.65 **(d)** 0.55

5. The game of Snakes and Ladders originated in ancient India from a game called: 1

(a) Mancala **(b)** Pachisi only **(c)** Shatranj **(d)** Jñān-Chaupaḍ

6. The 5th term of the GP with first term 2 and common ratio 3 is: 1

(a) 163 **(b)** 162 **(c)** 324 **(d)** 161

7. If $u_1 = 1$ and $u_n = 2u_{n-1} + 3$, then u_4 is: 1

(a) 15 **(b)** 27 **(c)** 13 **(d)** 29

8. The sequence $2, 5, 8, 11, \ldots$ is: 1

(a) an AP with $d = 2$ **(b)** a GP with $r = 3$

(c) neither an AP nor a GP **(d)** an AP with $d = 3$

9. The 10th triangular number is: 1

(a) 55 **(b)** 45 **(c)** 66 **(d)** 50

10. The Virahānka–Fibonacci sequence was first written down in connection with: 1

(a) the area of circles **(b)** calendar design

(c) Prakrit metre and poetry **(d)** planetary motion

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A coin is tossed 200 times and shows heads 112 times. Find the relative frequency of heads and compare it with $\frac{1}{2}$. 2

12. One month of the year is chosen at random. What is the probability that it has 31 days? 2

13. Is $1, -1, 1, -1, 1, \ldots$ a GP? Give the common ratio and the nth term. 2

14. How many black triangles are there at Stages 4 and 5 of the Sierpiński triangle? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A day of the week is chosen at random. What is the probability that it is a weekend day? Is a random pick of 'a day starting with T' more or less likely? 3

16. Show that the points $(1, 3), (2, 6), (3, 12)$ from the GP $3, 6, 12, \ldots$ do not lie on a straight line. 3

17. A coin was tossed 1000 times with 520 heads. Find the relative frequency of heads and the difference from the theoretical probability. 3

OR

One month is chosen at random. Find the probability that it has exactly 30 days. 3

18. Derive the formula for the sum of the first n natural numbers. 3

OR

Show that the sum of two consecutive triangular numbers is a perfect square; verify for t_{10} and t_9. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A card is drawn from a pack of 52. Let E be the event 'the card is a spade'. 5

(i) Find $P(E)$.

(ii) Find $P(\text{not } E)$ and show it equals $1 - P(E)$.

(iii) Find the probability of 'a spade or a heart'.

(iv) Find the probability of 'an ace or a king'.

OR

Case study: A class of 50 students gives its mode of travel to school: bus 18, bicycle 10, walking 12, car 10. 5

(i) Find the probability that a randomly chosen student cycles.

(ii) Find the probability for each mode and check that they add to 1.

(iii) Among 1200 students in the school, estimate the number who walk.

(iv) Which mode is most likely and by how much more than the least likely?

20. (i) Show that $25 + 26 + \cdots + 58 = 1411$ using $S_{58} - S_{24}$. 5

(ii) Check the answer using 'average of first and last × number of terms'.

(iii) Who first recorded this result and where?

OR

(i) Prove that $t_n + t_{n-1} = n^2$ for triangular numbers. 5

(ii) Verify for $n = 10$.

(iii) Find the 17th and 80th triangular numbers.

END OF QUESTION PAPER — Total: 40 Marks